DeepWatch: A Unified Deepfake Forensics and Attribution Platform

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Executive Summary

Deepfakes — AI-generated fake faces, voices, and videos — have become a tool of everyday crime, from voice-clone scam calls to fabricated video evidence. Existing research tackles only fragments of the problem: detection without explanation, attribution only for faces, and no system that produces evidence a victim can actually use. This report analyses four verified IEEE/CVF base papers, identifies their concrete limitations, and proposes **DeepWatch**: a unified platform that detects fakes, attributes them to their generation method, explains the evidence visually, and produces a forensic report — closing gaps that the current literature leaves open.

1. The Problem

Generative AI has made synthetic media cheap and convincing. Real incidents now include scam calls using cloned voices of family members, fake videos of public figures, and manipulated visual evidence. Victims and small investigators have no accessible tool to prove that a piece of media is fake: existing detectors output a bare score with no explanation, work only on clean laboratory data, and fail on media compressed by social platforms. The problem is therefore not only technical but practical — authenticity cannot be *demonstrated* by an ordinary person when it matters most.

2. Research Objectives

This work aims to (1) study the current state of deepfake detection and attribution as published in credible venues; (2) identify the specific, measurable limitations of the strongest existing methods; and (3) design a system that closes those gaps as an integrated, demonstrable platform, with improvements that can be verified against the base papers' published numbers.

3. Methodology

Recent literature (2022–2026) was surveyed through the arXiv open repository, with venue verification via the OpenAlex and Crossref scholarly APIs, since IEEE Xplore restricts automated access. Four papers in verified IEEE venues were selected as the base literature and read in detail for their methods, datasets, results, and stated or observable limitations. The proposed design was then derived directly from the identified gaps, so that every planned feature answers a documented weakness rather than adding novelty for its own sake.

4. Background and Base Papers

Deepfake research divides into detection (is it fake?), attribution (which method made it?), and speech anti-spoofing. The table below summarises the four base papers and their key limitations as published.

Base paper (venue, year)	Contribution	Key limitation
UCF, Yan et al., ICCV 2023 [1]	Generalizable detection via "common forgery features"; 0.996 AUC in-dataset	Drops to 0.828 AUC cross-dataset; needs forgery-method labels; no explanation
Up-sampling artifacts, Tan et al., CVPR 2024 [2]	Detects CNN-generator up-sampling artifacts	Blind to diffusion-based fakes (no such artifacts)
OW-DFA / CPL, Sun et al., ICCV 2023 [3]	Open-world attribution of unseen forgery methods	Face-only; low accuracy on unseen attacks; field still young
ASVspoof 2021, Liu et al., IEEE/ACM TASLP 2023 [4]	Benchmark for spoofed/deepfake speech	Audio-only; never fused with video analysis

5. Analysis: The Gap in Current Work

Read together, the base papers reveal a structural gap. Detection, attribution, and audio analysis exist as isolated research tracks; none of the reviewed systems combines them, and none produces human-usable evidence. Furthermore, the strongest detector (UCF) loses roughly a fifth of its accuracy on unseen data [1], the most cited artifact method is blind to diffusion generators [2], and attribution remains limited to narrow face-swap settings [3]. These are not marginal issues — they are the difference between a laboratory metric and a tool that works on a real scam video call.

6. Proposed Work: What We Will Newly Implement

DeepWatch is designed to close the documented gaps through five concrete innovations:

(a) Diffusion-aware detection — extend detection beyond GAN artifacts to diffusion-generated fakes, addressing the blind spot of [2]. **(b) Audio–video fusion** — jointly analyse face, voice, and lip-sync consistency in video calls, merging the video track of [1] with the speech track of [4] for the first time in a single pipeline. **(c) Unified attribution** — build on the open-world approach of [3] to name the likely generation method, extended beyond face-only forgeries. **(d) Explainable evidence** — visual heatmaps and plain-language reasons instead of a bare score. **(e) One-click forensic report** — a tamper-evident PDF (optionally hash-anchored on a test blockchain) that a victim can present to authorities — a capability absent from all reviewed systems. Improvements will be measured against the base papers' published metrics (AUC, attribution accuracy) so the contribution is verifiable.

7. Challenges

Cross-dataset generalization remains the hardest open problem in the field, so improvement targets will be modest and honestly reported. Diffusion-era datasets are newer and less standardized than FaceForensics++. Social-media compression destroys low-level artifacts, requiring compression-robust training. Finally, fusing audio and video raises synchronization and compute demands that must be managed within a student project's resources.

8. Recommendations and Conclusion

It is recommended to adopt UCF [1] and OW-DFA [3] as the twin technical bases, prioritise innovations (a) and (b) as the core contribution, and deliver (d) and (e) as the differentiating features. The conclusion of this research is that deepfake forensics is a real, current, and under-served problem; that credible IEEE/CVF foundations exist but leave clear, documented gaps; and that a unified detect–attribute–explain–report platform is both feasible and novel. DeepWatch is therefore proposed for implementation.

References

[1] Z. Yan, Y. Zhang, Y. Fan, B. Wu, "UCF: Uncovering Common Features for Generalizable Deepfake Detection," IEEE/CVF ICCV 2023. DOI: 10.1109/ICCV51070.2023.02048 — arxiv.org/abs/2304.13949

[2] C. Tan et al., "Rethinking the Up-Sampling Operations in CNN-Based Generative Network for Generalizable Deepfake Detection," IEEE/CVF CVPR 2024 — indexed via api.openalex.org

[3] Z. Sun et al., "Contrastive Pseudo Learning for Open-World DeepFake Attribution," IEEE/CVF ICCV 2023 — arxiv.org/abs/2309.11132

[4] X. Liu et al., "ASVspoof 2021: Towards Spoofed and Deepfake Speech Detection in the Wild," IEEE/ACM Trans. Audio, Speech, and Language Processing, 2023 — indexed via api.openalex.org